Modern Networking
Unit 1 – Modern Networking

Chapter 2 :Requirements and Technology

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Unit	Details
I	Modern Networking Elements of Modern Networking The Networking Ecosystem ,Example Network Architectures,Global Network Architecture,A Typical Network Hierarchy Ethernet Applications of Ethernet Standards Ethernet Data Rates Wi-Fi Applications of Wi-Fi,Standards Wi-Fi Data Rates 4G/5G Cellular First Generation Second Generation, Third Generation Fourth Generation Fifth Generation, Cloud Computing Cloud Computing Concepts The Benefits of Cloud Computing Cloud Networking Cloud Storage, Internet of Things Things on the Internet of Things, Evolution Layers of the Internet of Things, Network Convergence Unified Communications, Requirements and Technology Types of Network and Internet Traffic,Elastic Traffic,Inelastic Traffic, Real-Time Traffic Characteristics Demand: Big Data, Cloud Computing, and Mobile TrafficBig Data Cloud Computing,,Mobile Traffic, Requirements: QoS and QoE,,Quality of Service,Quality of Experience, Routing Characteristics, Packet Forwarding, Congestion Control ,Effects of Congestion,Congestion Control Techniques, SDN and NFV Software-Defined Networking,Network Functions Virtualization Modern Networking Elements

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PART I MODERN NETWORKING

Chapter 1: Elements of Modern Networking

1.1 The Networking Ecosystem
1.2 Example Network Architectures
1.3 Ethernet
1.4 Wi-Fi
1.5 4G/5G Cellular
1.6 Cloud Computing
1.7 Internet of Things
1.8 Network Convergence
1.9 Unified Communications

Chapter 2: Requirements and Technology

2.1 Types of Network and Internet Traffic
2.2 Demand: Big Data, Cloud Computing
2.3 Requirements: QoS and QoE
2.4 Routing
2.5 Congestion Control
2.6 SDN and NFV
2.7 Modern Networking Elements

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Introduction to chapter

- Networks will make possible many straightforward and significant economies.
- There will be problems such as loss of control, a potential lack of responsiveness to changing needs, and priority conflicts; but many of these problems have already been solved to a considerable degree.
- Chapter 1, "Elements of Modern Networking," provided a survey of the elements that make up the networking ecosystem, including network technologies, network architecture, services, and applications. In a concise fashion, this chapter provides motivation, technical background, and an overview of the key topics covered in

- **Chapter Objectives**:
- After studying this chapter, you should be able to
- Present an overview of the major categories of packet traffic on the Internet and internets, including elastic, inelastic, and real-time traffic.
- Discuss the traffic demands placed on contemporary networks by big data, cloud computing, and mobile traffic.
- Explain the concept of quality of service.
- Explain the concept of quality of experience

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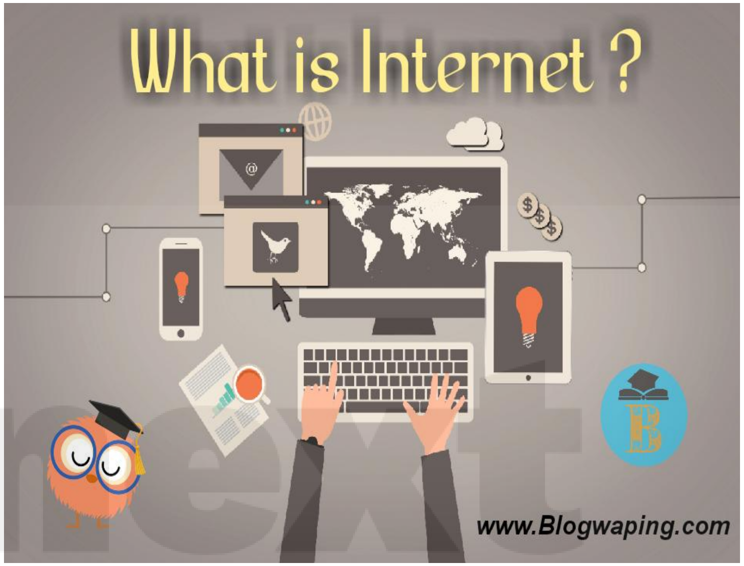
TYPES OF NETWORK AND INTERNET TRAFFIC

- **Internet :**
- A worldwide internetwork based on TCP/IP that interconnects thousands of public and private networks and millions of users. internet (with lower case i)
- A large network made up of a number of smaller networks. Also referred to as an internetwork.
- Traffic on the Internet and enterprise networks can be divided into two broad categories:
- <u>elastic and inelastic</u>.
- A consideration of their differing requirements clarifies the need for an enhanced networking architecture.



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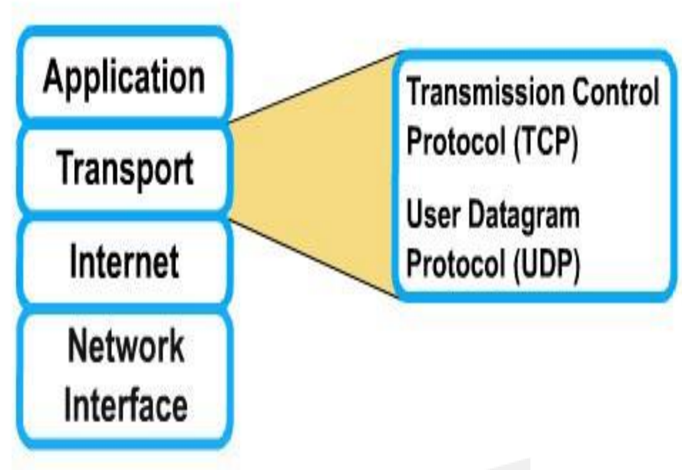
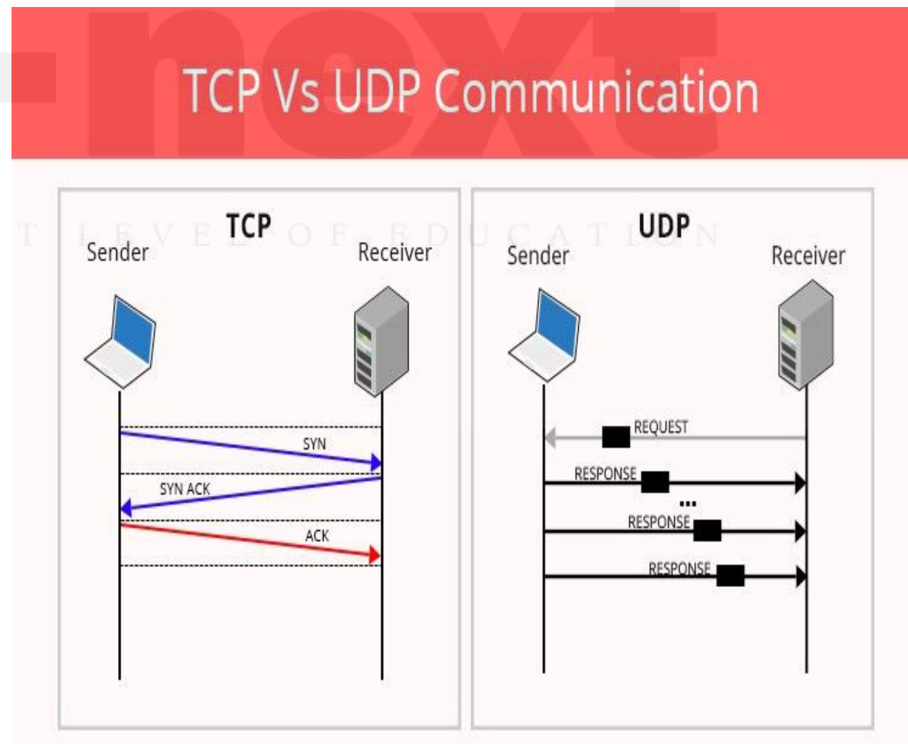
Elastic Traffic

- Elastic traffic is that which can adjust, over wide ranges, to changes in delay and throughput across an internet and still meet the needs of its applications.
- This is the traditional type of traffic supported on TCP/IP-based internets and is the type of traffic for which internets were designed.
- Applications that generate such traffic typically use Transmission Control Protocol (TCP) or User Datagram Protocol (UDP) as a transport protocol.
- In the case of UDP, the application will use as much capacity as is available up to the rate that the application generates data.
- In the case of TCP, the application will use as much capacity as is available up to the maximum rate that the end-toend receiver can accept data.





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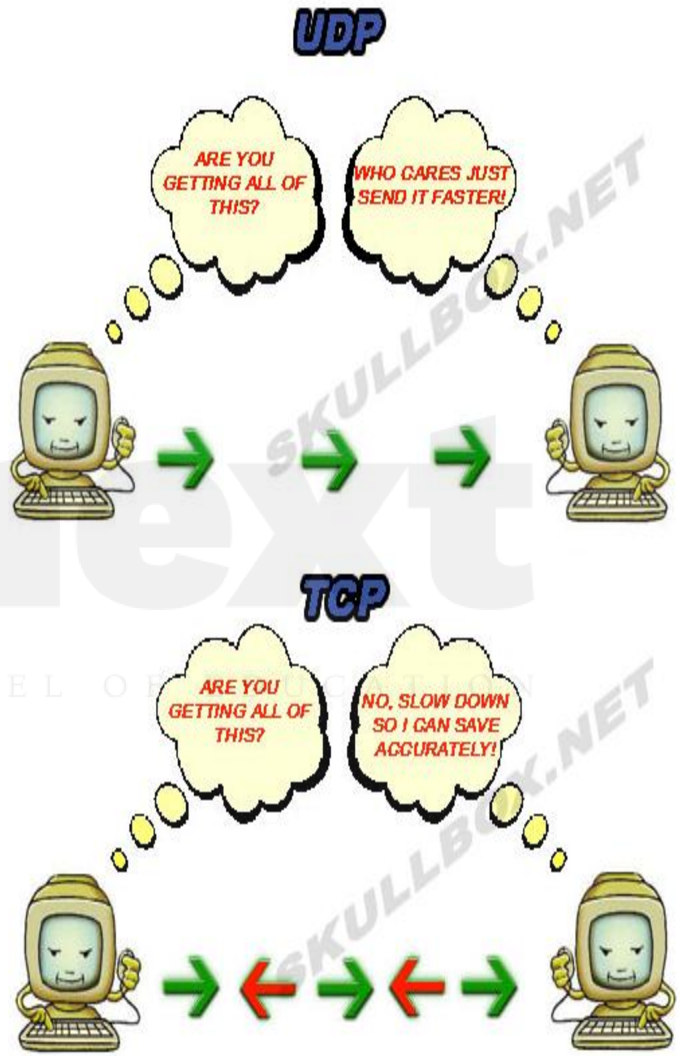
Elastic Traffic

- Applications that can be classified as elastic include the common applications that operate over TCP or UDP, including file transfer (File Transfer Protocol / Secure FTP [FTP/SFTP]), electronic mail (Simple Mail Transport Protocol [SMTP]), remote login (Telnet, Secure Shell [SSH]), network management (SimpleNetwork Management Protocol [SNMP]), and web access (Hypertext Transfer Protocol / HTTP Secure [HTTP/HTTPS]). However, there are differences among the requirements of these applications, including the following:
- E-mail is generally insensitive to changes in delay.
- When file transfer is done via user command rather than as an automated background task, the user expects the delay to be proportional to the file size and so is sensitive to changes in throughput.
- With network management, delay is generally not a serious concern. However, if failures in an internet are the cause of congestion, then the need for SNMP messages to get through with minimum delay increases with increased congestion



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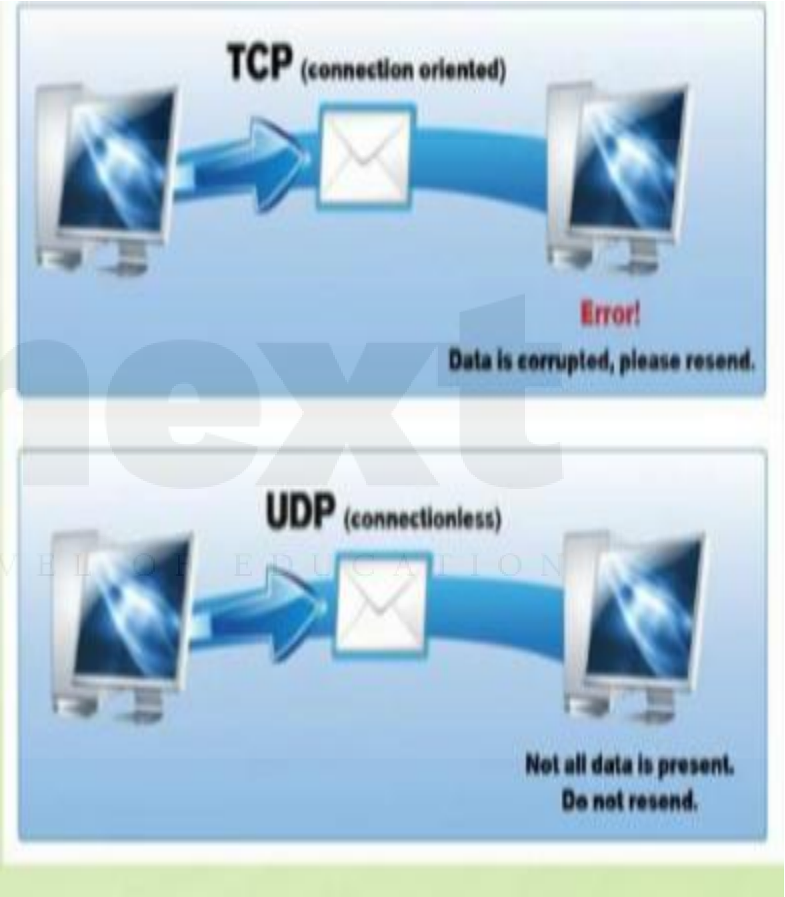
Elastic Traffic

- It is important to realize that it is not per packet delay that is the quantity of interest.
- Observation of real delays across the Internet suggest that wide variations in delay do not occur.
- Because of the congestion control mechanisms in TCP, when congestion develops, delays only increase modestly before the arrival rate from the various TCP connections slow down.
- Instead, the quality of service (QoS) perceived by the user relates to the total elapsed time to transfer an element of the current application.
- For an interactive Telnet-based application, the element may be a single keystroke or single line.
- For web access, the element is a web page, which could be as little as a few kilobytes or could be substantially larger for an image-rich page.
- For a scientific application, the element could be many megabytes of data.



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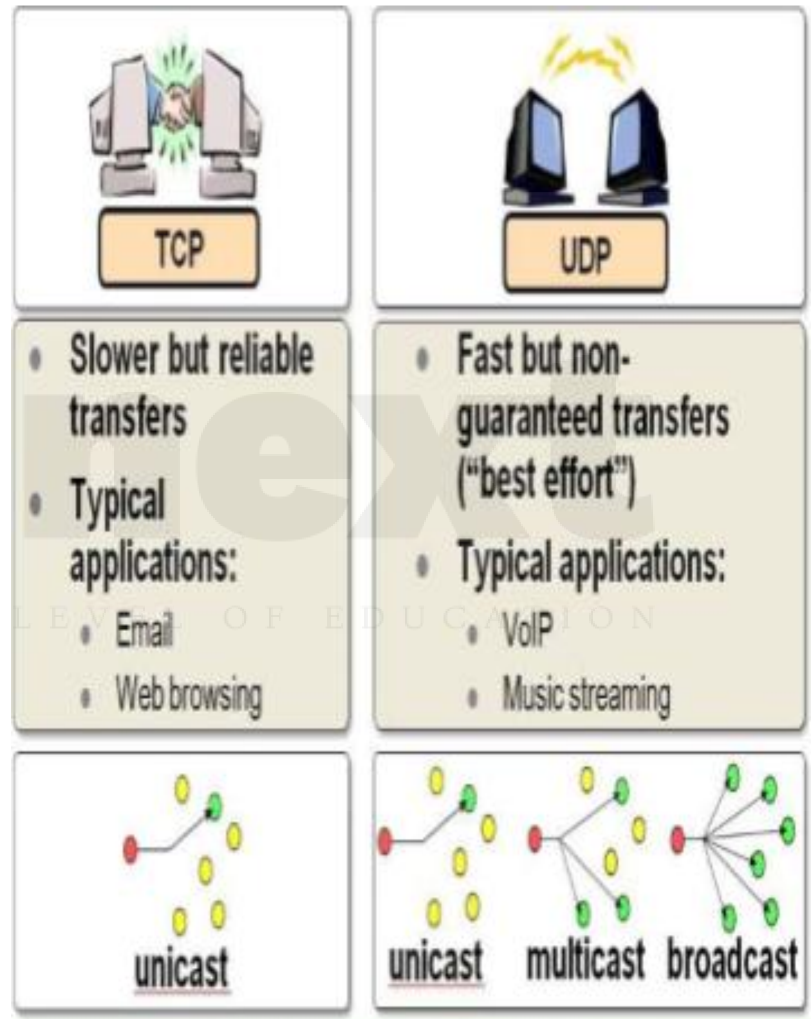
Elastic Traffic

- For very small elements, the total elapsed time is dominated by the delay time across the Internet.
- However, for larger elements, the total elapsed time is dictated by the sliding-window performance of TCP and is therefore dominated by the throughput achieved over the TCP connection.
- Thus, for large transfers, the transfer time is proportional to the size of the file and the degree to which the source slows because of congestion.
- It should be clear that even if you confine your attention to elastic traffic, some service prioritizing and controlling traffic could be of benefit.



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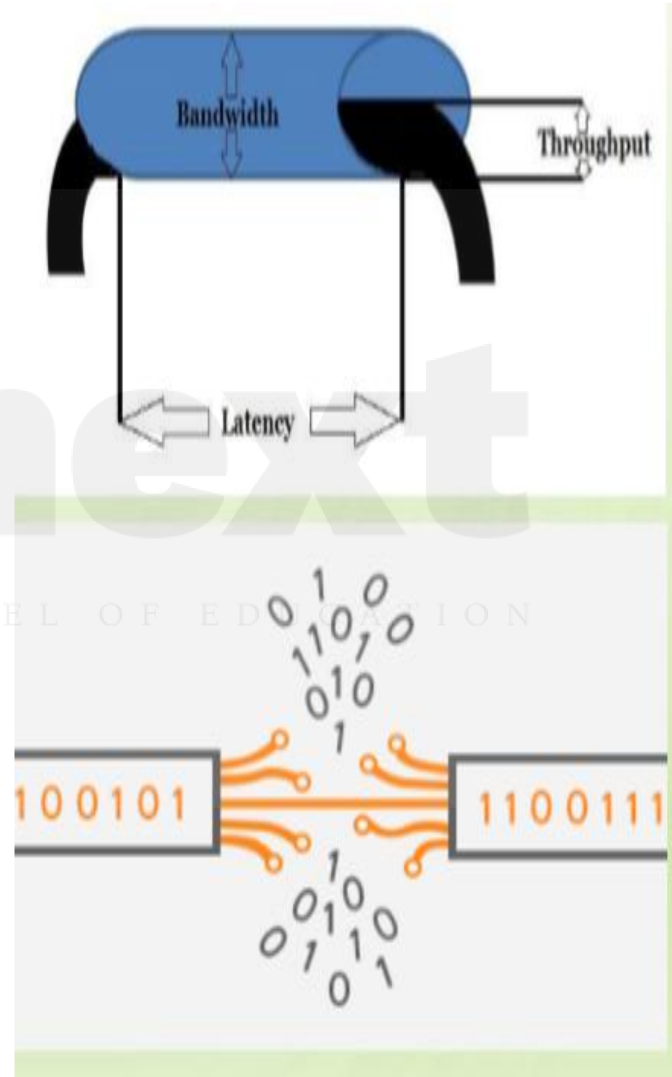
Inelastic Traffic

- does not easily adapt, if at all, to changes in delay and throughput across an internet.
- **Examples of inelastic traffic include multimedia transmission, such as voice and video, and high-volume interactive traffic, such as an interactive simulation application (for example, airline pilot simulation).**
- The requirements for inelastic traffic may include the following:
- **Throughput:** A minimum throughput value may be required. Unlike most elastic traffic, which can continue to deliver data with perhaps degraded service, many inelastic applications absolutely require a given minimum throughput.
- **Delay**: Also called latency. An example of a delay-sensitive application is stock trading; someone who consistently receives later service will consistently act later, and with



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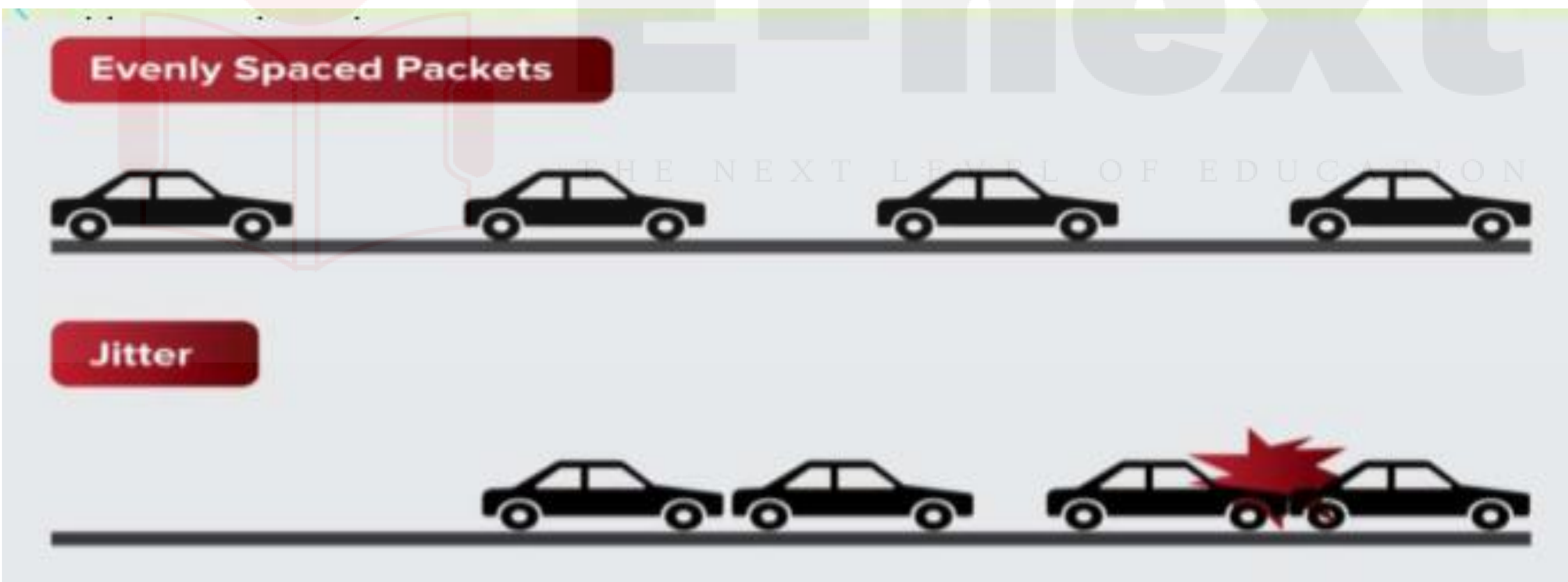
- **Delay jitter:** The magnitude of delay variation, called delay jitter, or simply jitter, is a critical factor in real-time applications.
- Because of the variable delay imposed by an internet, the interarrival times between packets are not maintained at a fixed interval at the destination. To compensate for this, the incoming packets are buffered, delayed sufficiently to compensate for the jitter, and then released at a constant rate to the software that is expecting a steady real-time stream.
- The larger the allowable delay variation, the longer the real delay in delivering the data and the greater the size of the delay buffer required at receivers. Real-time interactive applications, such as teleconferencing, may require a reasonable upper bound on jitter



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Inelastic Traffic

- These requirements are difficult to meet in an environment with variable queuing delays and congestion losses.
- Accordingly, inelastic traffic introduces two new requirements into the internet architecture.
- First, some means is needed to give preferential treatment to applications with more demanding requirements.
- Applications need to be able to state their requirements, either ahead of time in some sort of service request function, or on the fly, by means of fields in the IP packet header.
- The former approach provides more flexibility in stating requirements, and it enables the network to anticipate demands and deny new requests if the required resources are unavailable.
- This approach implies the use of some sort of resource reservation protocol.

Voice	One-way latency ≤ 150 ms
	One-way peak-to-peak jitter ≤ 30 ms
	Per-hop peak-to-peak jitter ≤ 10 ms
	Packet loss ≤ 1 percent
Broadcast video	Packet loss ≤ 0.1 percent
Real-time interactive video	One-way latency ≤ 200 ms
	One-way peak-to-peak jitter ≤ 50 ms
	Per-hop peak-to-peak jitter ≤ 10 ms
	Packet loss ≤ 0.1 percent
Multimedia conferencing	One-way latency ≤ 200 ms
	Packet loss ≤ 1 percent
Multimedia streaming	One-way latency ≤ 400 ms
	Packet loss ≤ 1 percent

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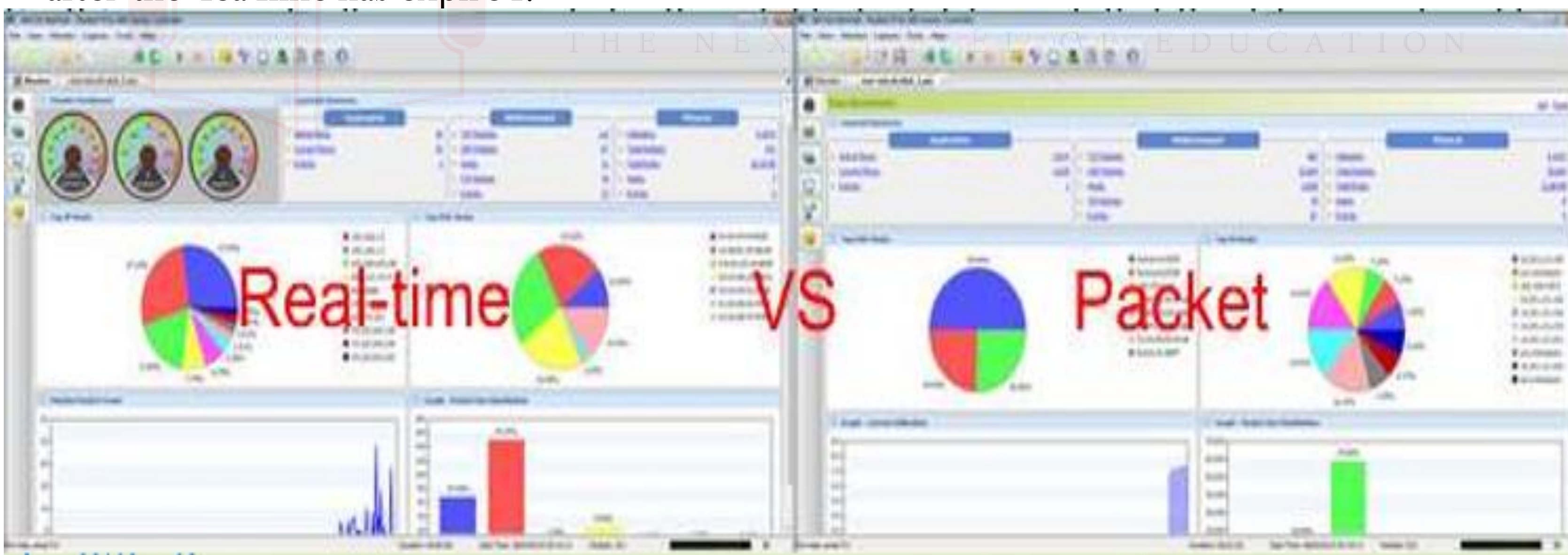
Real-Time Traffic Characteristics

- As mentioned, a common example of inelastic traffic is Realtime traffic.
- With traditional elastic applications, such as file transfer, electronic mail, and client/server applications including the web, the performance metrics of interest are generally throughput and delay.
- There is also a concern with reliability, and mechanisms are used to make sure that no data are lost, corrupted, or misordered during transit.
- By contrast, realtime applications are concerned with timing issues as well as packet loss.
- In most cases, there is a requirement that data be delivered at a constant rate equal to the sending rate.
- In other cases, a deadline is associated with each block of data, such that the data are not usable after the deadline has expired.



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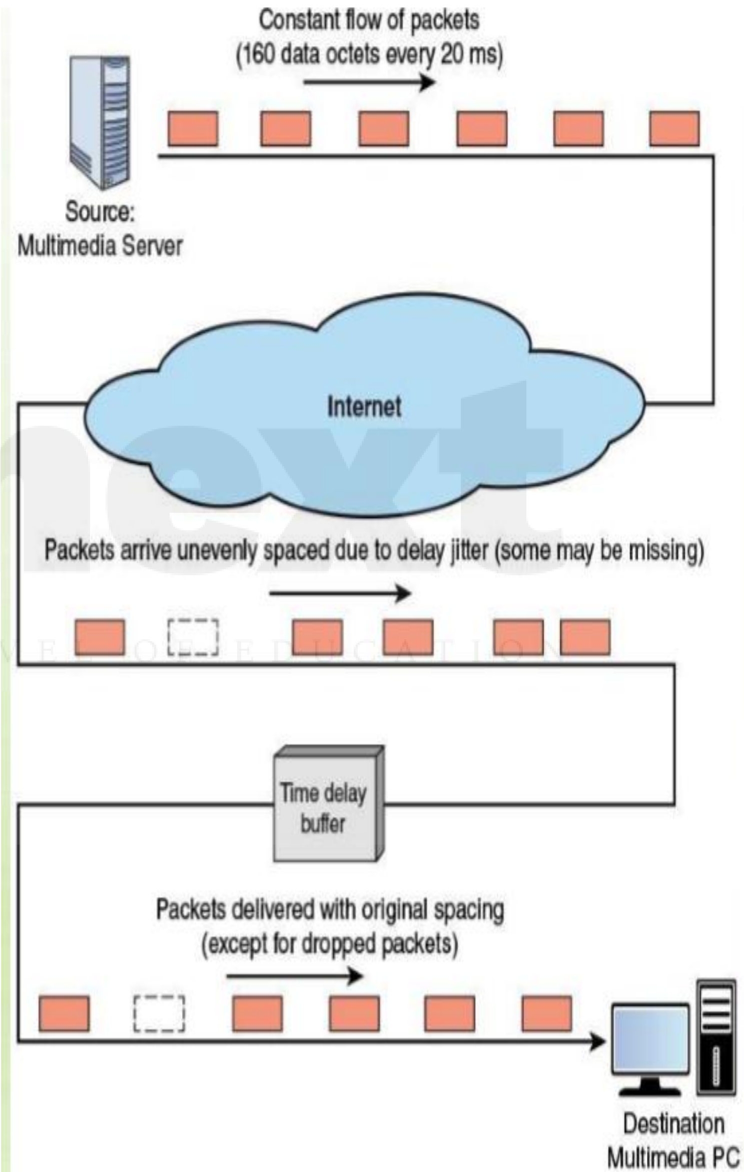
Real-Time Traffic Characteristics

- Figure illustrates a typical real-time environment , Here, a server is generating audio to be transmitted at 64 kbps.
- The digitized audio is transmitted in packets containing 160 octets of data, so that one packet is issued every 20 ms.
- These packets are passed through an internet and delivered to a multimedia PC, which plays the audio in real time as it arrives.
- However, because of the variable delay imposed by the internet, the interarrival times between packets are not maintained at a fixed 20 ms at the destination.
- To compensate for this, the incoming packets are buffered, delayed slightly, and then released at a constant rate to the software that generates the audio.
- The buffer may be internal to the destination machine or in an external network device.
- The compensation provided by the delay buffer is limited.



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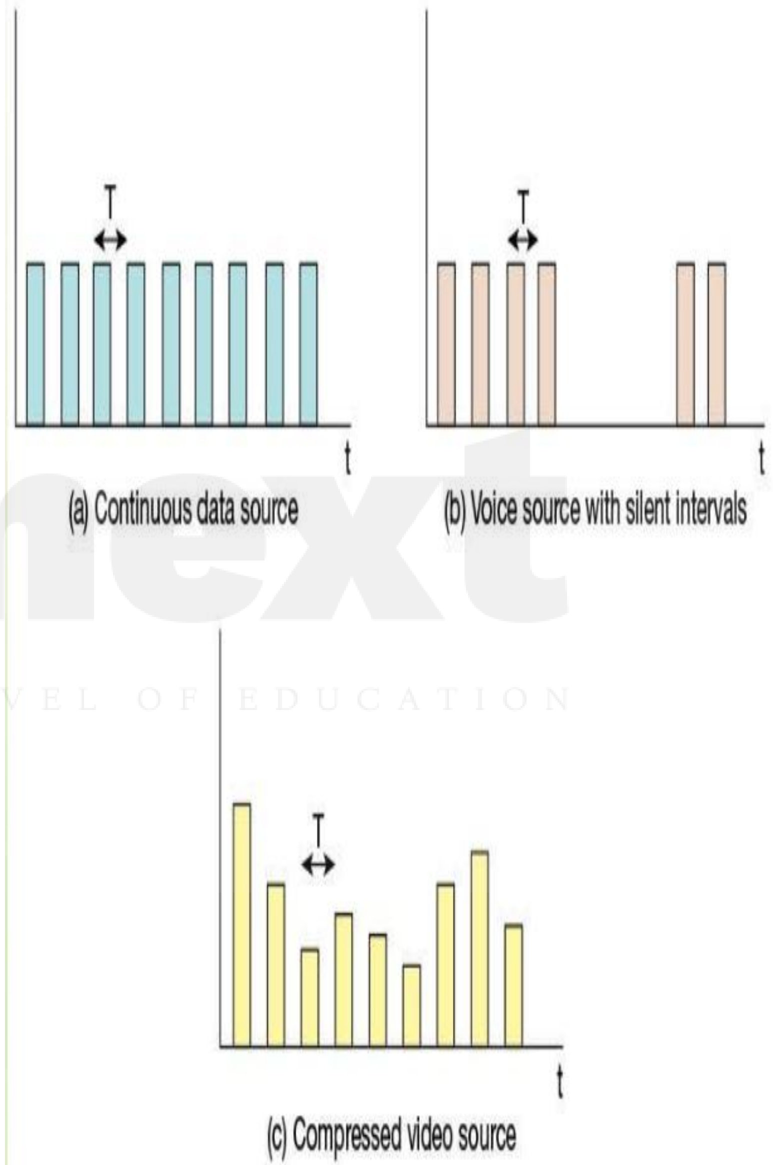
Real-Time Traffic Characteristics:

- The description of real-time traffic so far implies a series of equal-size packets generated at a constant rate. This is not always the profile of the traffic ,Figure illustrates some of the common possibilities, as described in the list that follows.
- **Continuous data source:** Fixed-size packets are generated at fixed intervals. This characterizes applications that are constantly generating data, have few redundancies, and that are too important to compress in a lossy way. Examples are air traffic control radar and real-time simulations. Control radar and realtime simulations.
- **On/off source:** The source alternates between periods when fixed-size packets are generated at fixed intervals and periods of inactivity. A voice source, such as in telephony or audio conferencing, fits this profile.
- **Variable packet size:** The source generates variable-length packets at uniform intervals. An example is digitized video in which



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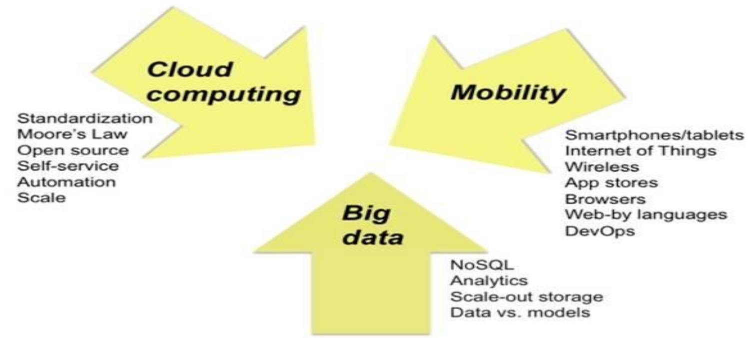
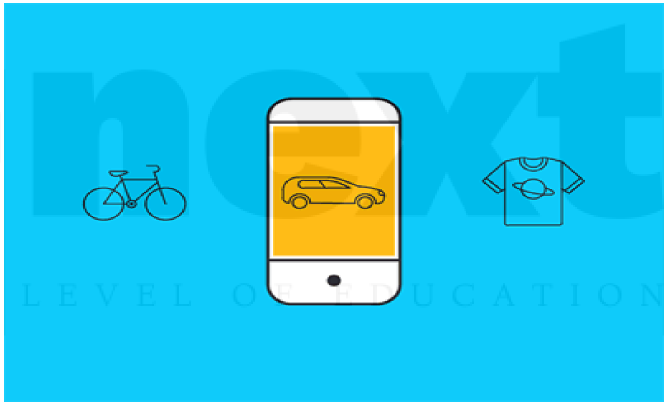
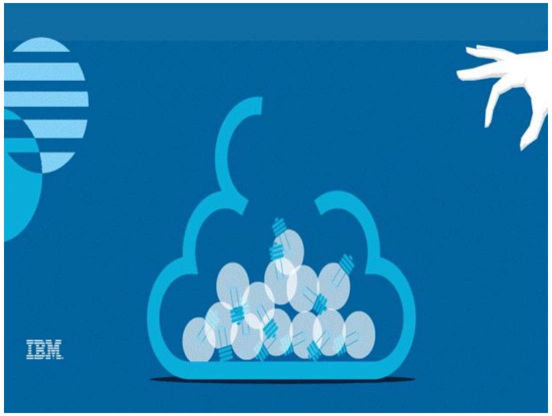
DEMAND: BIG DATA, CLOUD COMPUTING, AND MOBILE TRAFFIC

- Having looked at the types of traffic presented to the Internet and other IP-based networks, consider the application areas that are generating the greatest stress on network resources and management.
- Three areas stand out: `
  - big data,
  - cloud computing
  - mobility.







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Big Data

- A collection of data on such a large scale that standard data analysis and management tools are not adequate.
- In simple terms, big data refers to everything that enables an organization to create, manipulate, and manage very large data sets (measured in terabytes, petabytes, exabytes, and so on) and the facilities in which these are stored.
- Distributed data centers, data warehouses, and cloudbased storage are common aspects of today's enterprise networks.
- Many factors have contributed to the merging of "big data" and business networks, including continuing declines in storage costs, the maturation of data mining and business intelligence (BI) tools, and government regulations and court cases that have caused organizations to stockpile large masses of structured and unstructured data, including documents, e-mail messages, voice-mail messages, text messages, and social media data.

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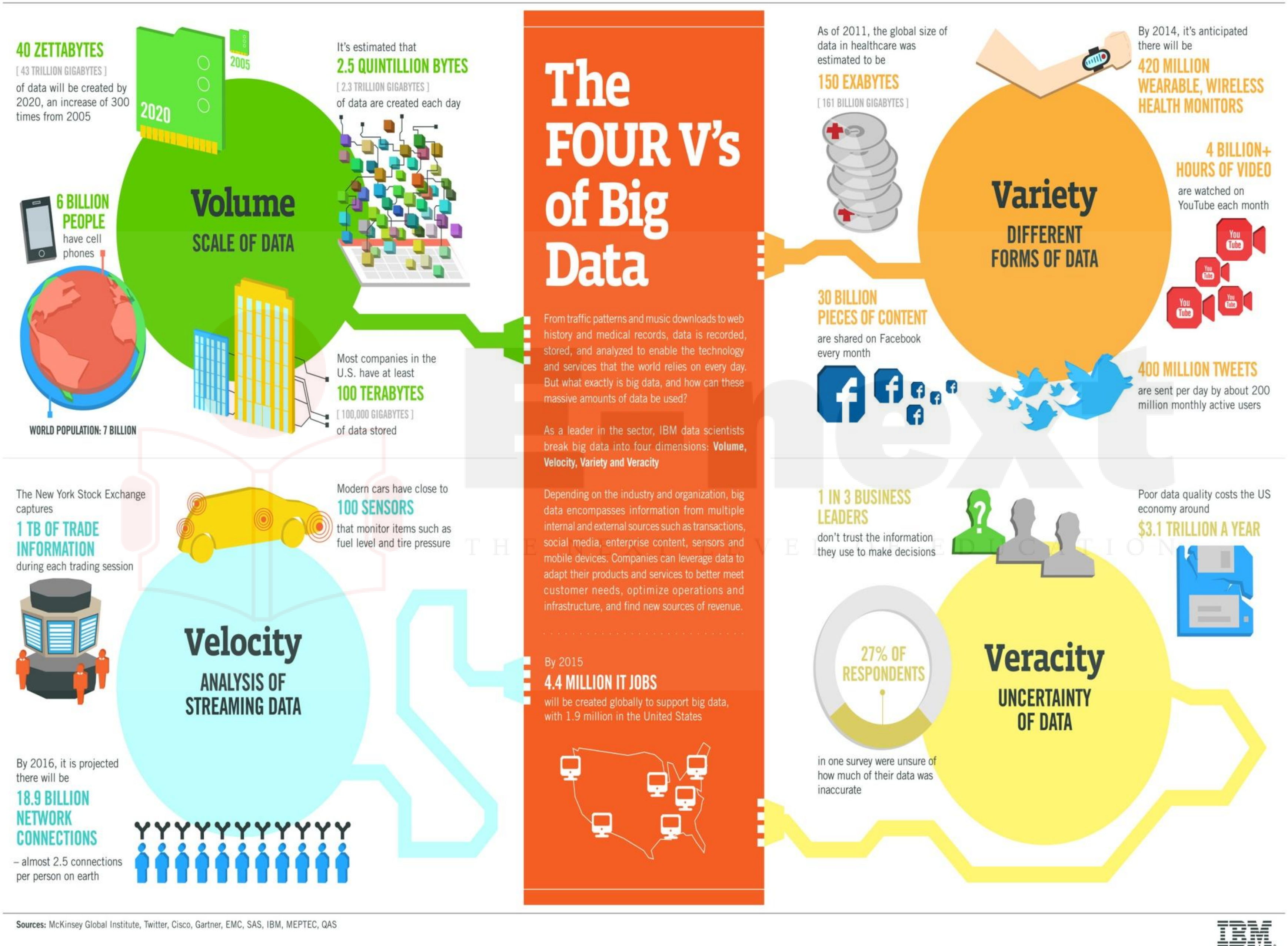

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Big Data Infrastructure Considerations :

- Traditional business data storage and management technologies include relational database management systems (RDBMS), network-attached storage (NAS), storage-area networks (SANs), data warehouses (DWs), and business intelligence (BI) analytics.
- Traditional data warehouse and BI analytics systems tend to be highly centralized within an enterprise infrastructure.
- These often include a central data repository with a RDBMS, high-performance storage, and analytics software, such as online analytical processing (OLAP) tools for mining and visualizing data.

Analytics :

- Analysis of massive amounts of data, particularly with a focus on decision making.
- Increasingly, big data applications are becoming a source of competitive value for businesses, especially those that aspire to build data products and services to profit from the huge volumes of data that they capture and store.

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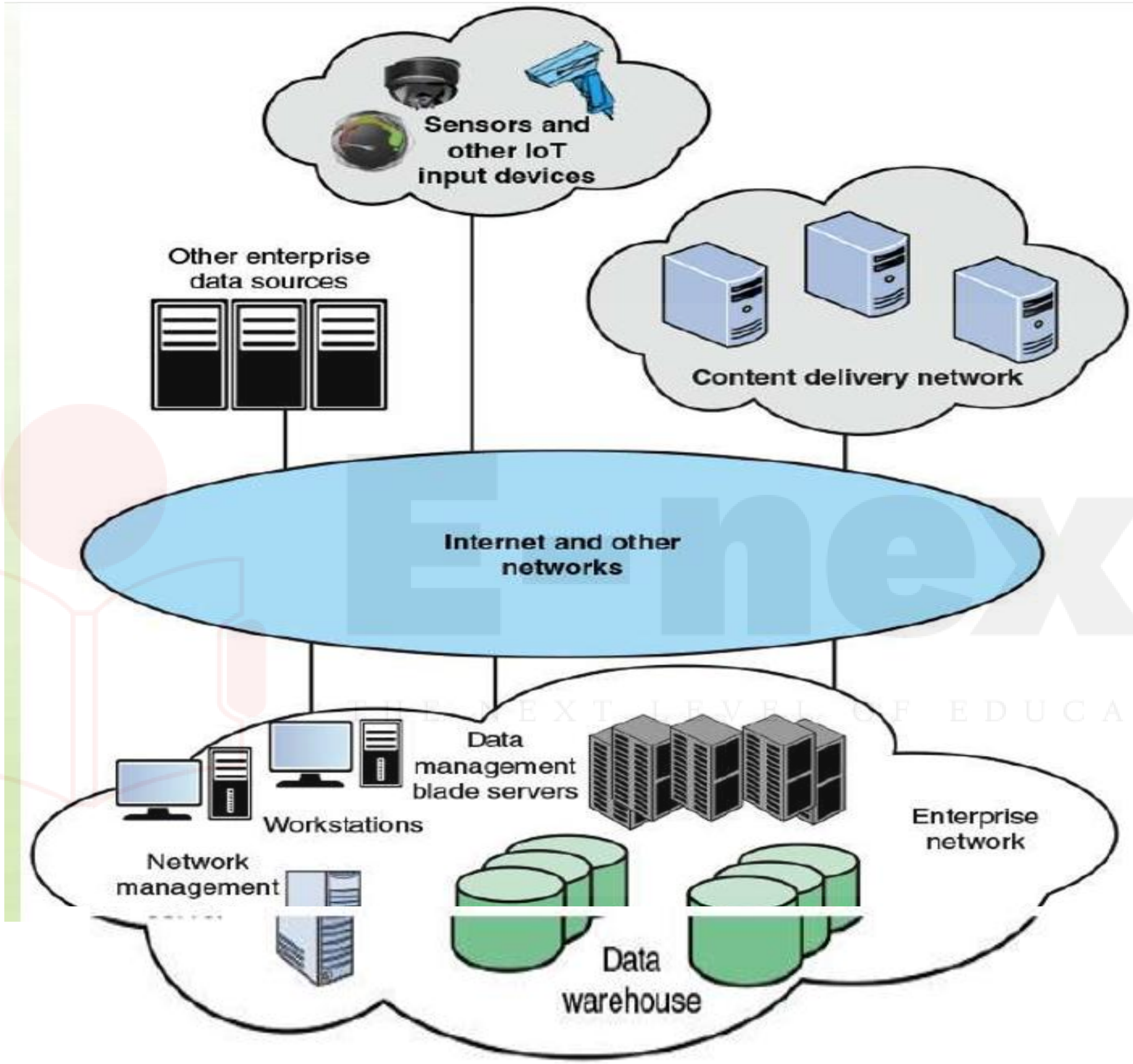

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Key elements within the enterprise include the following:

- **Data warehouse:** The DW holds integrated data from multiple data sources, used for reporting and data analysis.
- **Data management servers:** Large banks of servers serve multiple functions with respect to big data. The servers run data analysis applications, such as data integration tools and analytics tools. Other applications integrate and structure data from enterprise activity, such as financial data, point-of-sale data, and ecommerce activity.
- **Workstations / data processing systems:** Other systems involved in the use of big data applications and in the generation of input to big data warehouses.
- **Network management server:** One or more servers responsible for network management, control, and monitoring

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Key elements within the enterprise include the following:

- **Network capacity:** Running big data analytics requires a lot of capacity on its own; the issue is magnified when big data and day-to-day application traffic are combined over an enterprise network.
- **Latency:** The real or near-real-time nature of big data demands a network architecture with consistent low latency to achieve optimal performance.
- **Storage capacity:** Massive amounts of highly scalable storage are required to address the insatiable appetite of big data, yet these resources must be flexible enough to handle many different data formats and traffic loads.
- **Processing:** Big data can add significant pressure on computational, memory, and storage systems, which, if not properly addressed, can negatively impact operational efficiency.
- **Secure data access:** Big data combine sensitive data and provides data security

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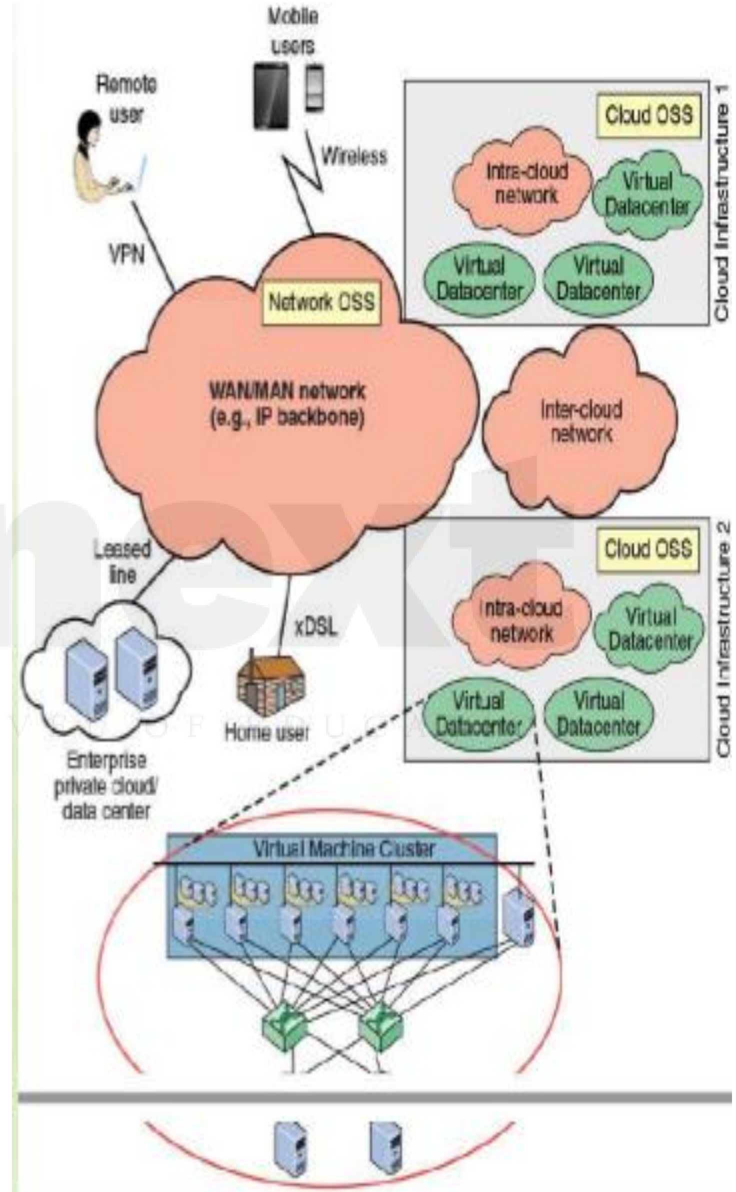
Cloud Computing

- As with big data installations, cloud computing presents imposing challenges for effective and efficient flow of traffic through networks.
- It will be helpful in this regard to consider the cloud network model developed by ITU-T, and shown in the Figure.
- This figure indicates the scope of network concerns for cloud network and service providers and for cloud service users.
- A cloud service provider maintains one or more local or regional cloud infrastructures.
- An intracloud network connects the elements of the infrastructure, including database servers, storage arrays, and other servers
- (for example, firewalls, load balancers, application acceleration devices, and IDS/IPS).
- The intracloud network will likely include a number of LANs interconnected with IP routers.



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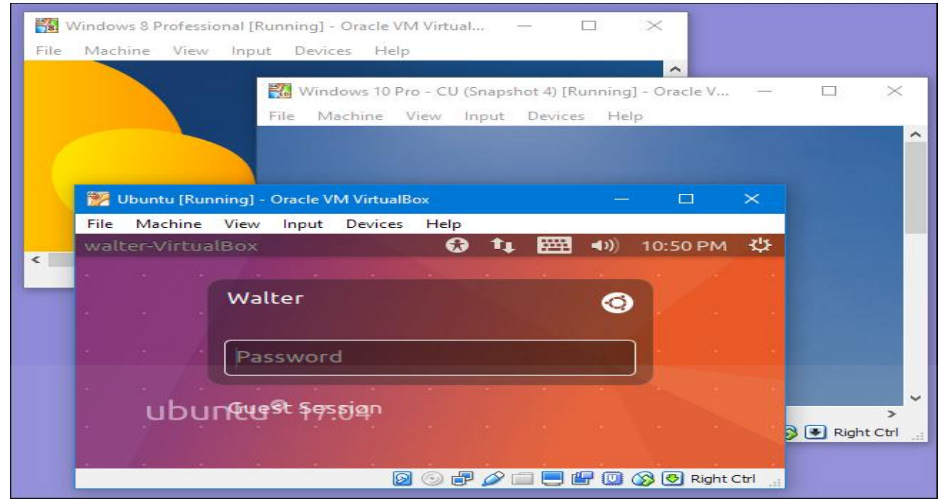
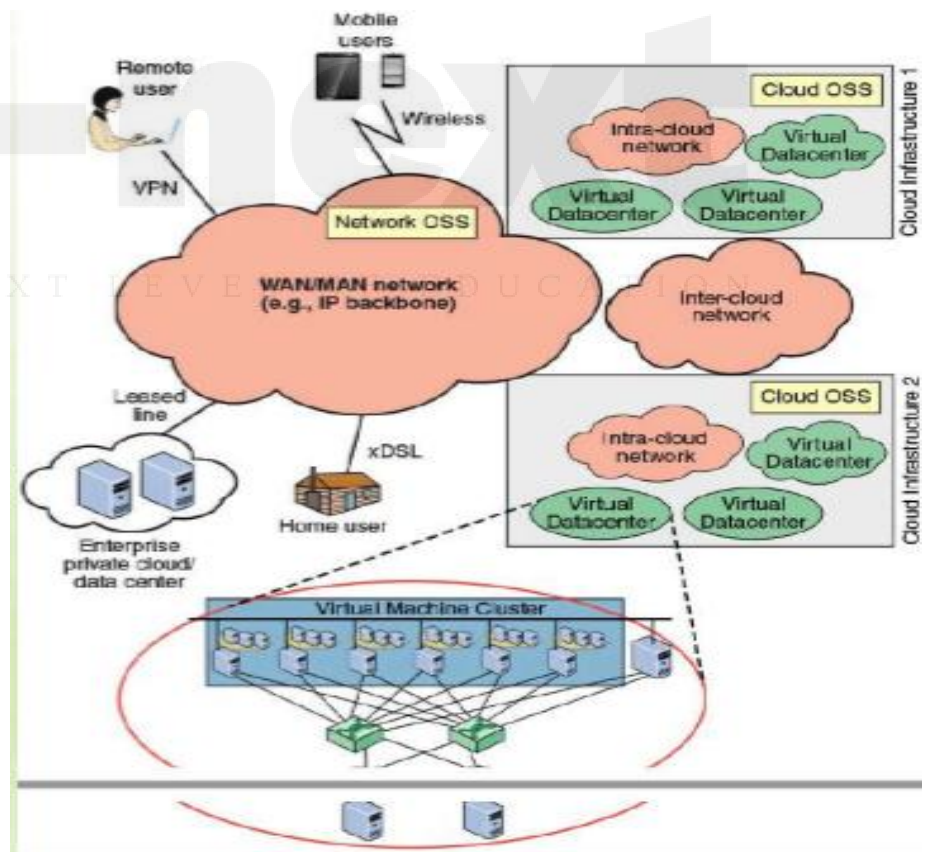
Cloud Computing

- **virtual machine :** One instance of an operating system along with one or more applications running in an isolated partition within the computer. It enables different operating systems to run in the same computer at the same time and prevents applications from interfering with each other.
- Intercloud networks interconnect cloud infrastructures together. These cloud infrastructures may be owned by the same cloud provider or by different ones.
- Finally, a core transport network is used by customers to access and consume cloud services deployed within the cloud provider's data center.





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lists the following functional requirements for this network capability:

Scalability:

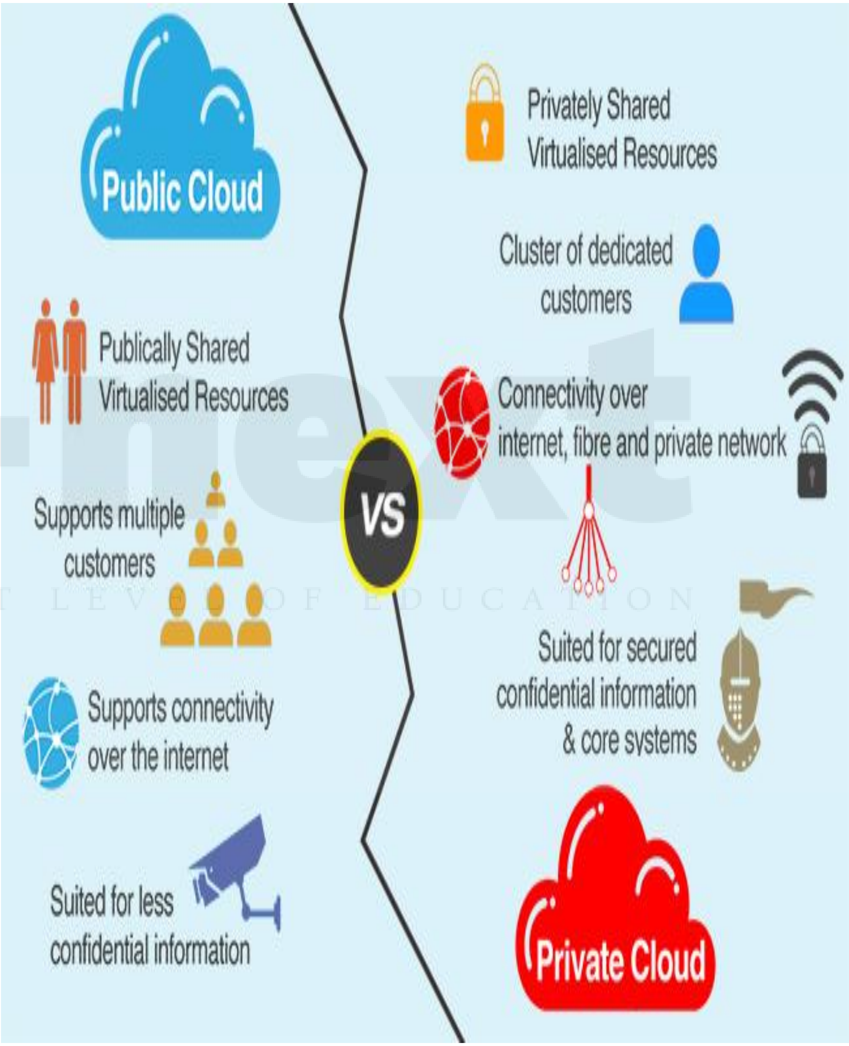
- Networks must be able to scale easily to meet the demands of moving from current cloud infrastructures of hundreds or a few thousand servers to networks of tens or even hundreds of thousands of servers.
- This scaling presents challenges in areas such as addressing, routing, and congestion control.

Agility and flexibility:

- The cloud-based data center needs to be able to respond and manage the highly dynamic nature of cloud resource utilization.
- This includes the ability to adapt to virtual machine mobility and to provide fine-grained control of flows routing through the data center.

Performance:

- Traffic in both big data installations and cloud provider networks is unpredictable and quite variable [KAND12].
- There are sustained spikes between nearby servers within the same rack and intermittent heavy traffic with a single source server and multiple destination servers.



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Mobile Traffic

- Technical innovations have contributed to the success of what were originally just mobile phones.
- The prevalence of the latest devices, with multimegabit Internet access, mobile apps, high megapixel digital cameras, access to multiple types of wireless networks (for example, Wi-Fi, Bluetooth, 3G, 4G), and several onboard sensors, all add to this momentous achievement.
- Devices have become increasingly powerful while staying easy to carry.
- Battery life has increased (even though device energy usage has also expanded), and digital technology has improved reception and allowed better use of a finite spectrum.
- As with many types of digital equipment, the costs associated with mobile devices have been decreasing.
- The first rush to wireless was for voice.

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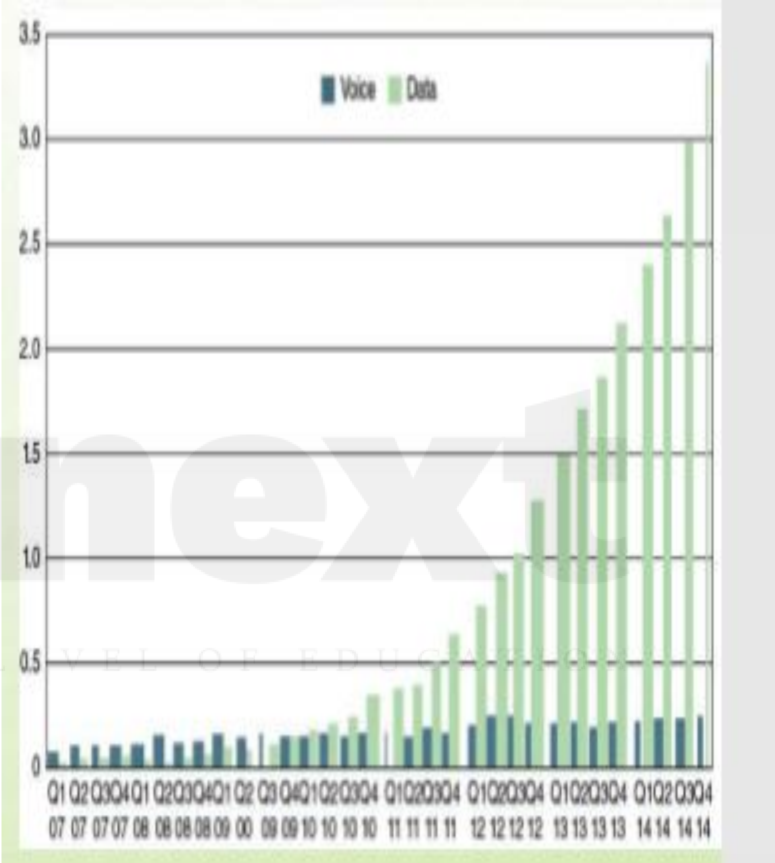
- **Mobile traffic :**
- The Figure shows the dramatic trend in world total mobile traffic in 2G, 3G, and 4G networks (not including DVB-H, Wi-Fi, and Mobile WiMAX) estimated by Ericsson [AKAM15].
- Ericsson's presence in more than 180 countries and its customer base representing more than 1000 networks enable it to measure mobile voice and data volumes.
- The result is a representative base for calculating world total mobile traffic.
- A big part of the mobile market is the wireless Internet.
- Wireless users use the Internet differently than fixed users, but in many ways no less effectively.
- Wireless smartphones have limited displays and input capabilities compared with larger devices such as laptops or PCs, but mobile apps give quick access to intended information without using websites.



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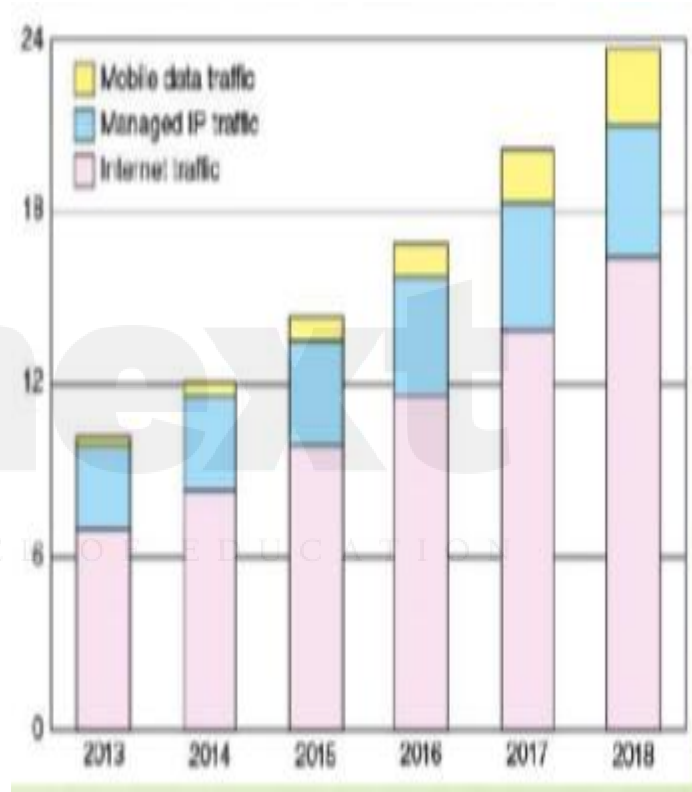
Mobile traffic

- The Figure shows a projection for mobile enterprise IP traffic, where the term enterprise refers to businesses and governments.
- Cisco's methodology rests on a combination of analyst projections, in-house estimates and forecasts, and direct data collection.
- As with mobile data traffic over cellular networks, mobile enterprise IP traffic is on a strong growth curve.
- The FIGURE Forecast Monthly Enterprise IP Traffic (Exabytes/Month)
- The Figure breaks the mobile traffic down into three categories:
  - **Mobile data traffic:** All enterprise traffic that crosses a mobile access point
  - **Managed IP traffic:** All enterprise traffic that is transported over IP but remains within the corporate WAN
  - **Internet traffic:** All enterprise traffic that crosses the public Internet Although mobile traffic is the smallest of the three categories of enterprise traffic, it is growing



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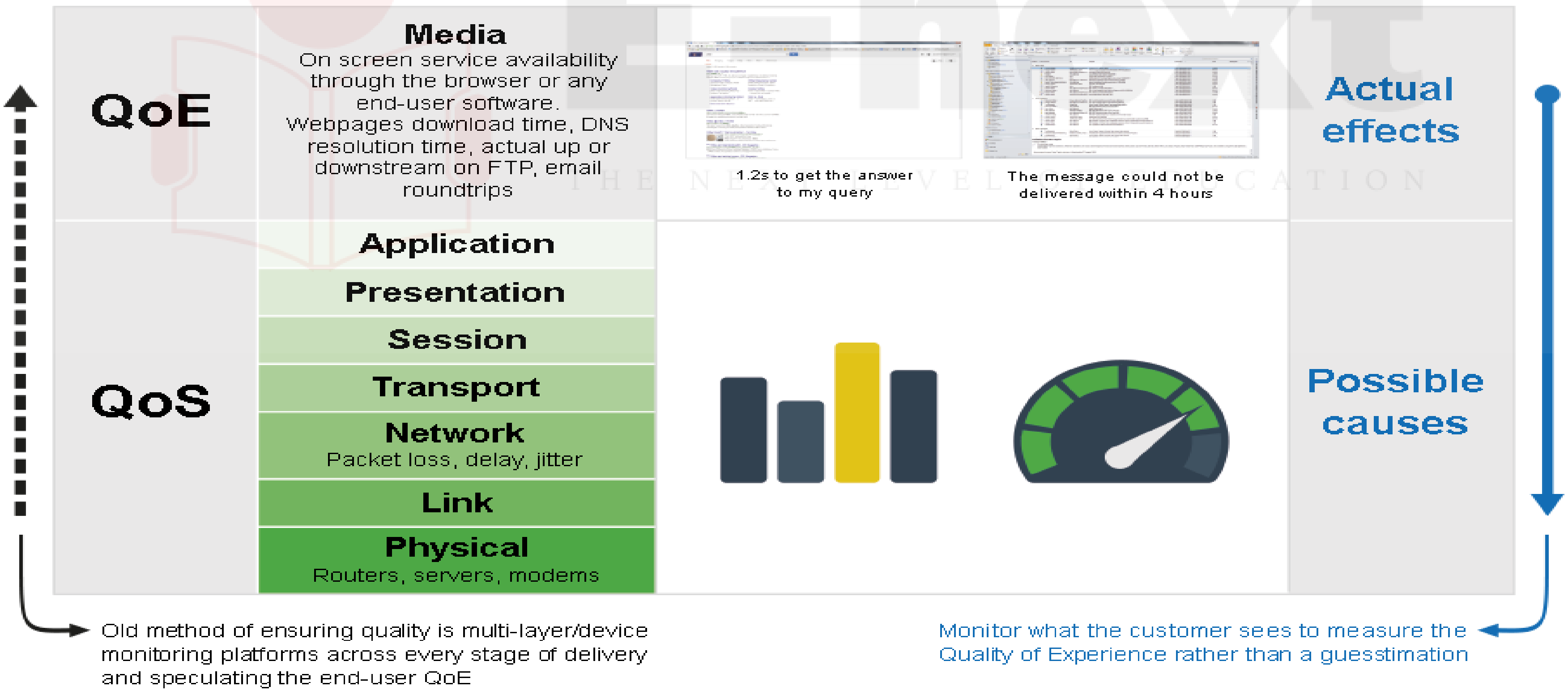
REQUIREMENTS: QOS AND QOE

- **quality of service (QoS) and quality of experience (QoE).**
- QoS and QoE enable the network manager to determine whether the network is meeting user needs and to diagnose problem areas that require adjustment to network management and network traffic control.



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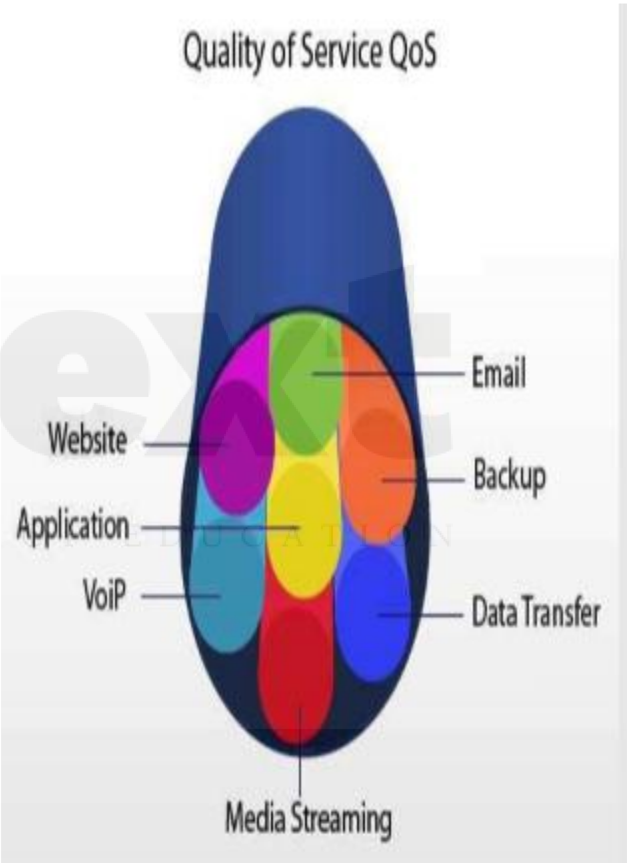
Quality of Service

- You can define QoS as the measurable end-to-end performance properties of a network service, which can be guaranteed in advance by a service level agreement (SLA) between a user and a service provider, so as to satisfy specific customer application requirements.

Commonly specified properties include the following:

- **Throughput:** A minimum or average throughput, in bytes per second or bits per second, for a given logical connection or traffic flow.
- **Delay:** The average or maximum delay. Also called latency.
- **Packet jitter**: Typically, the maximum allowable jitter.
- **Error rate**: Typically maximum error rate, in terms of fraction of bits delivered in error.
- **Packet loss:** Fraction of packets lost.
- **Priority:** A network may offer a given number of levels of priority. The assigned level for various traffic flows influences the way in which the different flows are handled by the network.
- **Availability:** Expressed as a percentage of time available.
- **Security**: Different levels or types of security may be defined.



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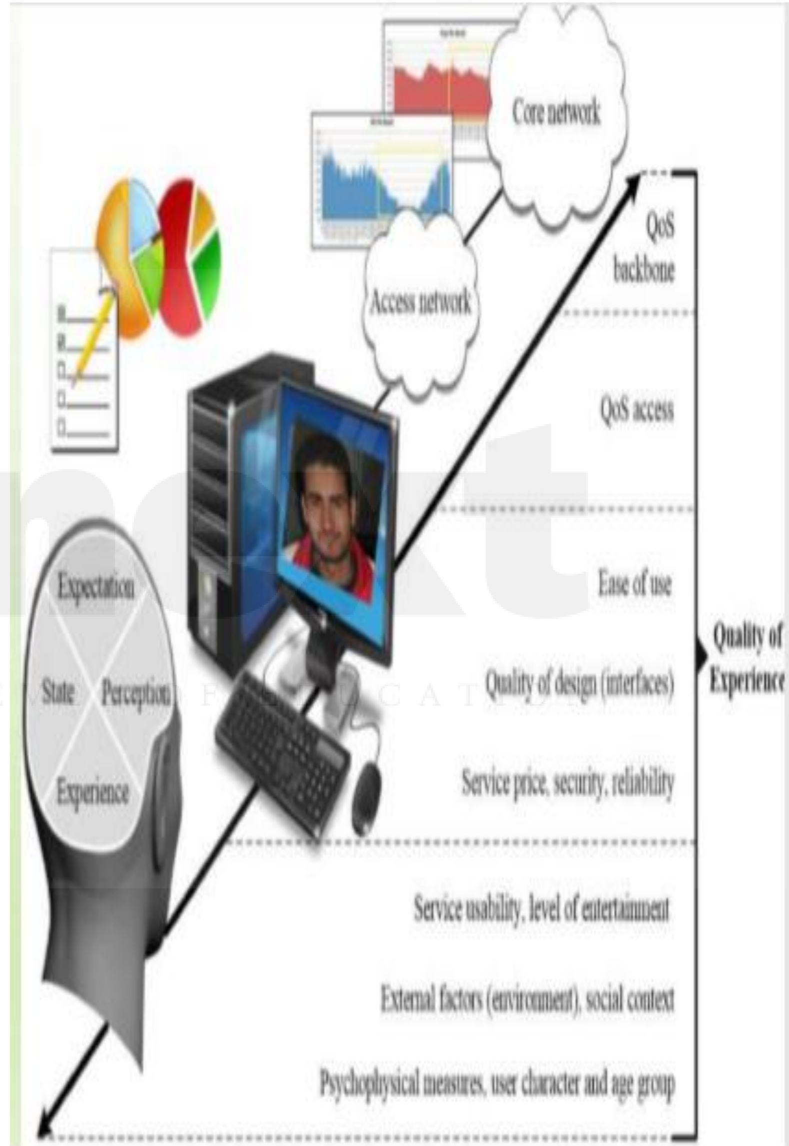
Quality of Experience

- QoE is a subjective measure of performance as reported by the user.
- Unlike QoS, which can be precisely measured, QoE relies on human opinion.
- QoE is important particularly when we deal with multimedia applications and multimedia content delivery.
- QoS provides measurable, quantitative targets that guide the design and operation of a network and enable customer and provider to agree on what quantitative performance the network will deliver for give applications and traffic flows.
- Although the maximum capacity may be fixed at a certain value by a media transmission system, this does not necessarily fix the quality of the multimedia content at, say, "high." This is because there are numerous ways the multimedia content could have been encoded, giving rise to differing perceived qualities.



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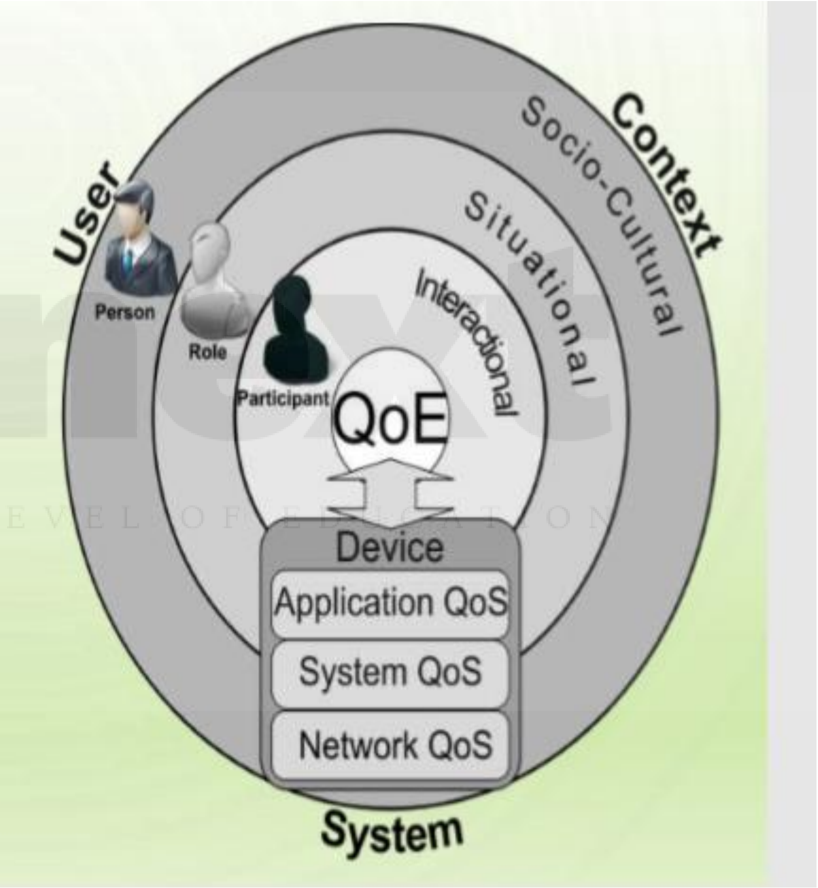
There is factors and features included in a requirement for QoE:

- **Perceptual:** This category encompasses the quality of the sensory aspects of the user experience. For video, examples include sharpness, brightness, contrast, flicker, and distortion. Audio examples include clarity and timbre.
- **Psychological:** This category deals with the user's feeling about the experience. Examples include ease of use, joy of use, usefulness, perceived quality, satisfaction, annoyance, and boredom.
- **Interactive:** This category deals with aspects of an experience related to the interaction between the user and the application or device, such as responsiveness, naturalness of interaction, communication efficiency and accessibility



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QOS or QOE ?

- For practical application, these last features need to be converted to quantitative measures.
- The management of QoE has become a crucial concept in the deployment of future successful applications, services, and products.
- The greatest challenges in providing QoE are developing effective methods for converting QoE features to quantitative measures and translating QoE measures to QoS measures.
- Whereas QoS can now easily be measured, monitored, and controlled at both the networking and application layers, and at both the end system and network sides, QoE is something that is still quite intricate to manage

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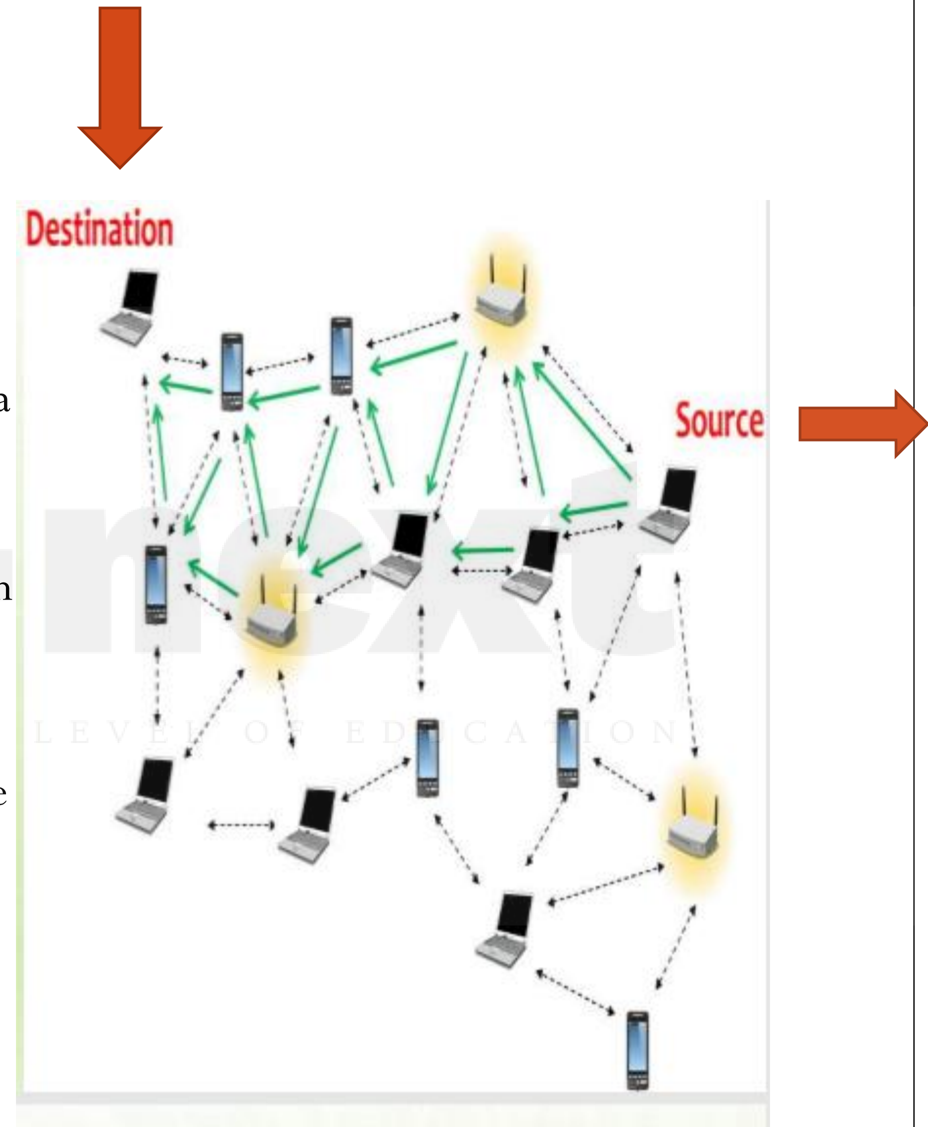
ROUTING

- The purpose here is to indicate the basic concepts of routing and congestion control, because these are the basic tools needed to support network traffic and to provide QoS and QoE.
- The primary function of an internet is to accept packets from a source station and deliver them to a destination station.
- To accomplish this, a path or route through the network must be determined; generally, more than one route is possible.
- Therefore, a routing function must be performed.
- The selection of a route is generally based on some performance criterion.
- The simplest criterion is to choose the minimum-hop route (one that passes through the least number of nodes) through the network.
- This is an easily measured criterion and should minimize the consumption of network resources.
- A generalization of the minimum-hop criterion is least-cost routing.



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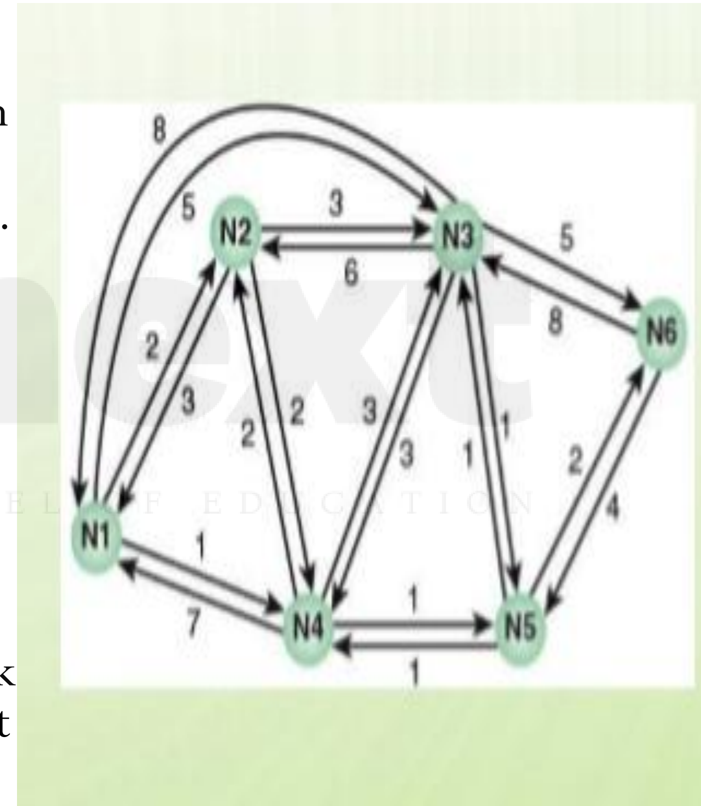
Routing

- Figure illustrates a network in which the two arrowed lines between a pair of nodes represent a link between these nodes, and the corresponding numbers represent the current link cost in each direction.
- Our concern, of course, is with an internet, in which each node is a router and the links between adjacent routers are networks or direct communications links.
- **The shortest path (fewest hops) from node 1 to node 6 is 1-3-6 (Cost = 5 + 5 = 10), but the least-cost path is 1-4-5-6 (Cost = 1 + 1 + 2 = 4).**
- Costs are assigned to links to support one or more design objectives.
- For example, the cost could be inversely related to the data rate (that is, the higher the data rate on a link the lower the assigned cost of the link) or the current link delay.
- In the first case, the least-cost route should provide the highest **throughput.**
- In the second case, the least-cost route should **minimize delay**



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Packet Forwarding

- The key function of any router is to accept incoming packets and forward them.
- For this purpose, a router maintains forwarding tables.
- Figure shows a simplified example of how this might be implemented for the network, with its associated link costs, of Figure
- A router's forwarding table shows, for each destination, the identity of the next node on the router.
- Each router may be responsible for discovering the appropriate routes.
- Alternatively, a network control center may be responsible for designing routes for all routers and maintaining a central forwarding table, providing each router with individual forwarding tables relevant only to that router.

CENTRAL FORWARDING TABLE

To Node \ From Node	1	2	3	4	5	6
1	–	1	5	2	4	5
2	2	–	5	2	4	5
3	4	3	–	5	3	5
4	4	4	5	–	4	5
5	4	4	5	5	–	5
6	4	4	5	5	6	–

Node 1 Table

Destination	Next Node
2	2
3	4
4	4
5	4
6	4

Node 2 Table

Destination	Next Node
1	1
3	3
4	4
5	4
6	4

Node 3 Table

Destination	Next Node
1	5
2	5
4	5
5	5
6	5

Node 4 Table

Destination	Next Node
1	2
2	2
3	5
5	5
6	5

Node 5 Table

Destination	Next Node
1	4
2	4
3	3
4	4
6	6

Node 6 Table

Destination	Next Node
[illegible]	5
2	5
3	5
4	5
5	5

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Packet Forwarding

- Note that it is not necessary to store the complete route for each possible pair of nodes. Rather, it is sufficient to know, for each pair of nodes, the identity of the first node on the route.
- In the Figure forwarding decisions are based solely on the identity of the destination system. Additional information is often used to determine the forwarding decision, such as the source address,
- **packet flow identifier, or security level of the packet:**
- **Failure:** When a node or link fails, it can no longer be used as part of a route.
- **Congestion:** When a particular portion of the network is heavily congested, it is desirable to route packets around rather than through the area of congestion.
- **Topology change:** The insertion of new links or nodes affects routing.

CENTRAL FORWARDING TABLE

From Node (columns) / To Node (rows)

To Node	1	2	3	4	5	6
1	–	1	5	2	4	5
2	2	–	5	2	4	5
3	4	3	–	5	3	5
4	4	4	5	–	4	5
5	4	4	5	5	–	5
6	4	4	5	5	6	–

Node 1 Table

Destination	Next Node
2	2
3	4
4	4
5	4
6	4

Node 2 Table

Destination	Next Node
1	1
3	3
4	4
5	4
6	4

Node 3 Table

Destination	Next Node
1	5
2	5
4	5
5	5
6	5

Node 4 Table

Destination	Next Node
1	2
2	2
3	5
5	5
6	5

Node 5 Table

Destination	Next Node
1	4
2	4
3	3
4	4
6	6

Node 6 Table

Destination	Next Node
[illegible]	5
2	5
3	5
4	5
5	5

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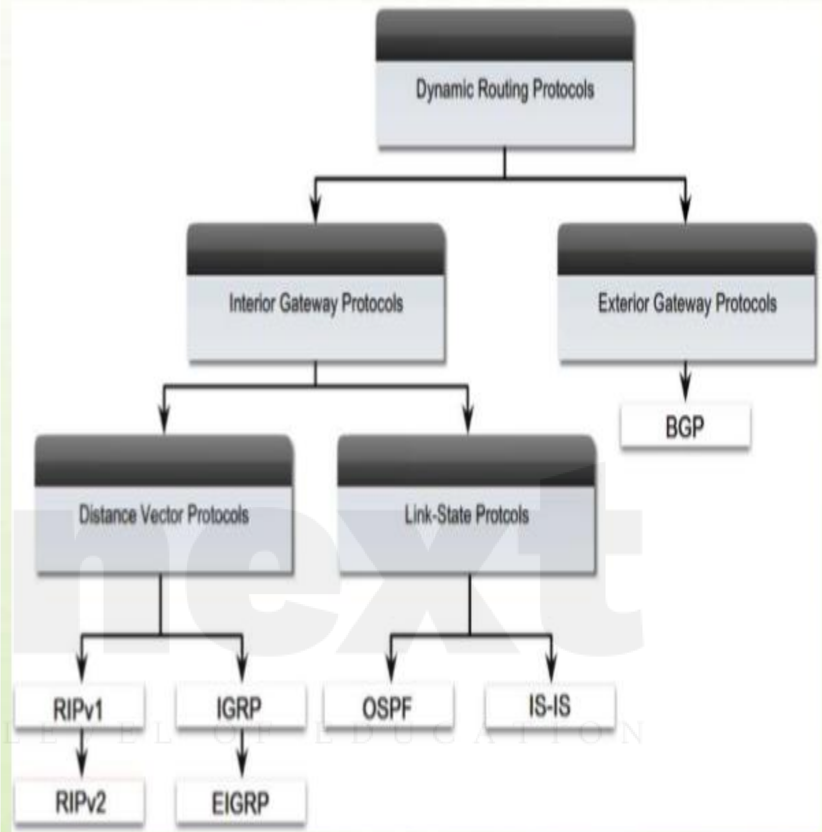
Routing Protocols

- The routers in an internet are responsible for receiving and forwarding packets through the interconnected set of networks.
- Each router makes routing decisions based on knowledge of the topology and traffic/delay conditions of the internet.
- Accordingly, a degree of dynamic cooperation is needed among the routers.
- In particular, the router must avoid portions of the network that have failed and should avoid portions of the network that are congested.
- To make such dynamic routing decisions, routers exchange routing information using a routing protocol.
- Information is needed about the status of the internet, in terms of which networks can be reached by which routes, and the delay characteristics of various routes.



In the literature, the terms interior gateway protocol (IGP) and exterior gateway protocol (EGP) are often used for what are referred to here as IRP and ERP. However, because the terms IGP and EGP also refer to specific protocols, we avoid their use when referring to the general concepts.

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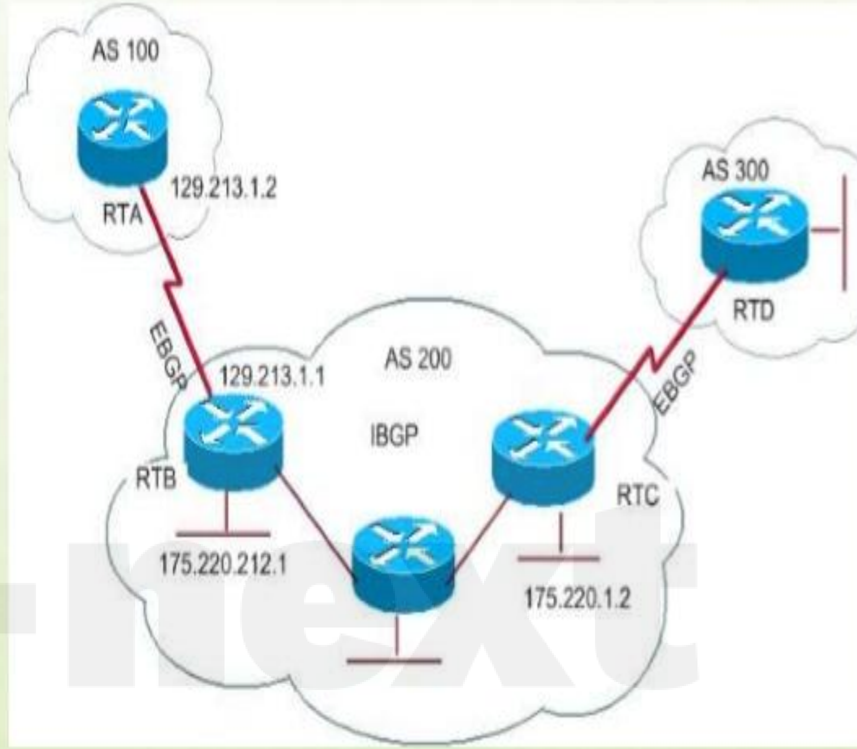
2.4 ROUTING

Routing Protocols

There are essentially two categories of routing protocols, which are based on the concept of an autonomous system (AS). We first define AS and then look at the two categories.

An AS exhibits the following characteristics:

1. An AS is a set of routers and networks managed by a single organization.
2. An AS consists of a group of routers exchanging information via a common routing protocol.
3. Except in times of failure, an AS is connected (in a graph-theoretic sense); that is, there is a path between any pair of nodes.



autonomous system (AS)

- A network that is administered by a single set of management rules that are controlled by one person, group, or organization. Autonomous systems often use only one routing protocol, although multiple protocols can be used. The core of the Internet is made up of many autonomous systems.

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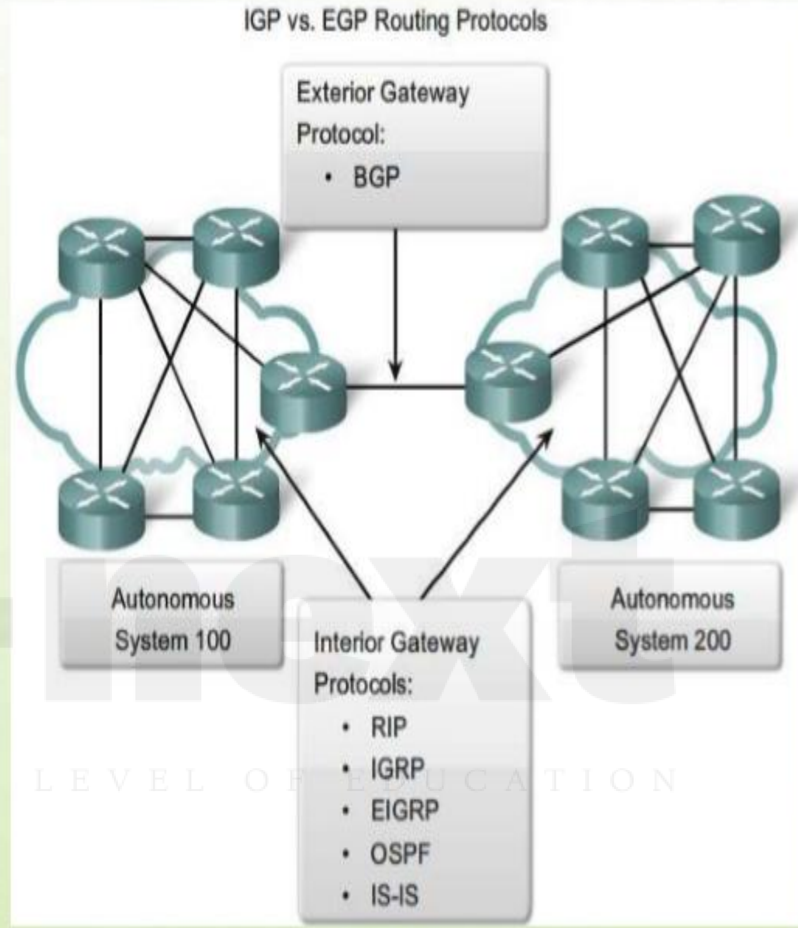
2.4 ROUTING

Routing Protocols

- A shared routing protocol, called here an interior router protocol (IRP), passes routing information between routers within an AS.
- The protocol used within the AS does not need to be implemented outside of the system.
- This flexibility allows IRPs to be custom tailored to specific applications and requirements.
- It may happen, however, that an internet will be constructed of more than one AS.
- For example, all the LANs at a site, such as an office complex or campus, could be linked by routers to form an AS.
- This system might be linked through a wide-area network to other autonomous systems.
- Figure illustrates this situation.



interior router protocol (IRP)

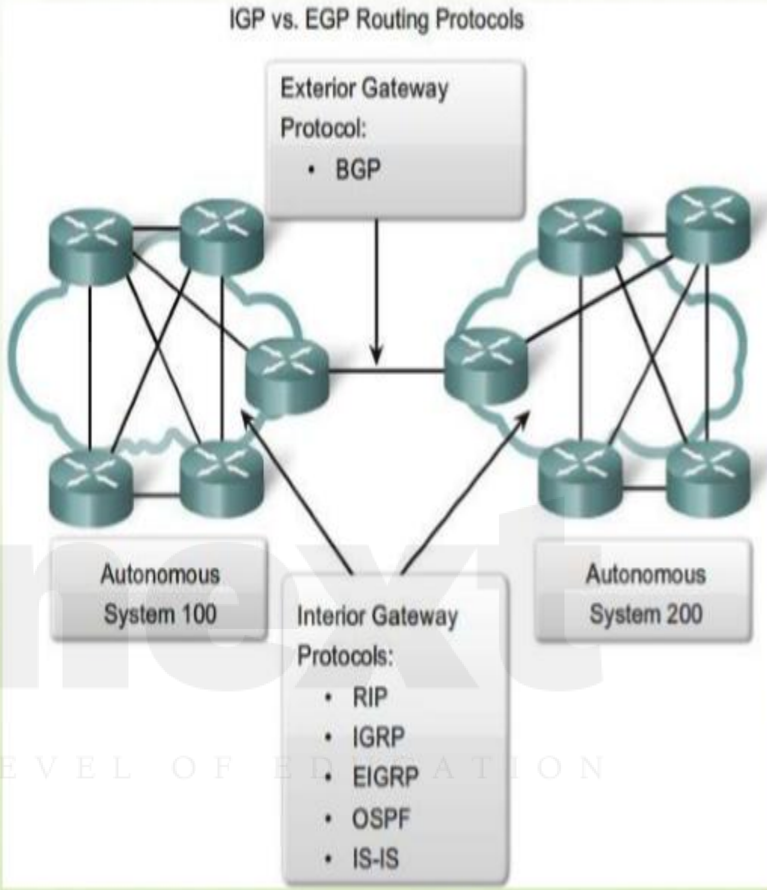
A protocol that distributes routing information to collaborating routers within an AS. Routing Information Protocol (RIP) and Open Shortest Path First (OSPF) Protocol are examples of IRPs. Historically, an IRP was referred to as an interior gateway protocol.

THE NEXT LEVEL OF EDUCATION

2.4 ROUTING

Routing Protocols

- In this case, the routing algorithms and information in routing tables used by routers in different autonomous systems may differ.
- Nevertheless, the routers in one AS need at least a minimal level of information concerning networks outside the system that can be reached.
- We refer to the protocol used to pass routing information between routers in different autonomous systems as an exterior router protocol (ERP).
- You can expect that an ERP will need to pass less information than an IRP for the following reason.
- If a packet is to be transferred from a host in one AS to a host in another AS, a router in the first system need only determine the target AS and devise a route to get into that target system.
- Once the packet enters the target AS, the routers within that system can cooperate to deliver the packet; the ERP is not concerned with, and does not know about, the details of the route followed within the target AS.



exterior router protocol (ERP)
A protocol that distributes routing information to collaborating routers that connect autonomous systems.
BGP is an example of an ERP. Historically, an ERP was referred to as an exterior gateway protocol.

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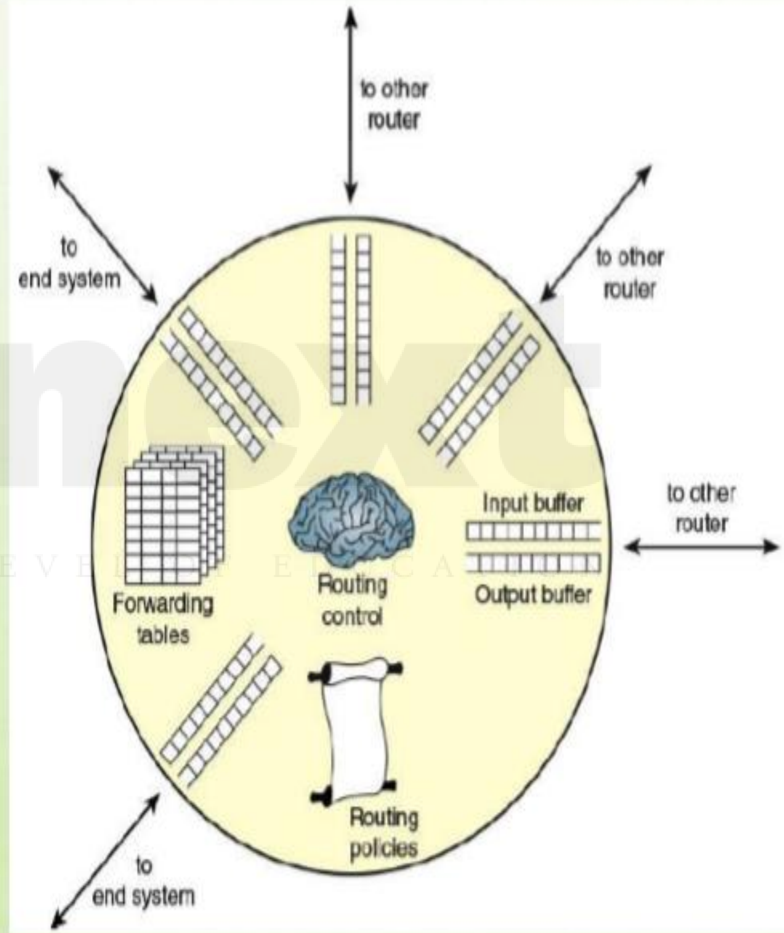
PART I MODERN NETWORKING

Chapter 2: Requirements and Technology

2.4 ROUTING

Elements of a Router

- Figure depicts the principal elements of a router, from the point of view of its routing function.
- Any given router has a number of I/O ports attached to it: one or more to other routers, and zero or more to end systems.
- On each port, packets arrive and depart.
- You can consider that there are two buffers, or queues, at each port: one to accept arriving packets, and one to hold packets that are waiting to depart.
- In practice,there might be two fixed-size buffers associated with each port, or there might be a pool of memory available for all buffering activities.
- In the latter case, you can think of each port having two variable-size buffers associated with it, subject to the constraint that the sum of all buffer sizes is a constant.
- In any case, as packets arrive, they are stored in the input buffer of the corresponding port.



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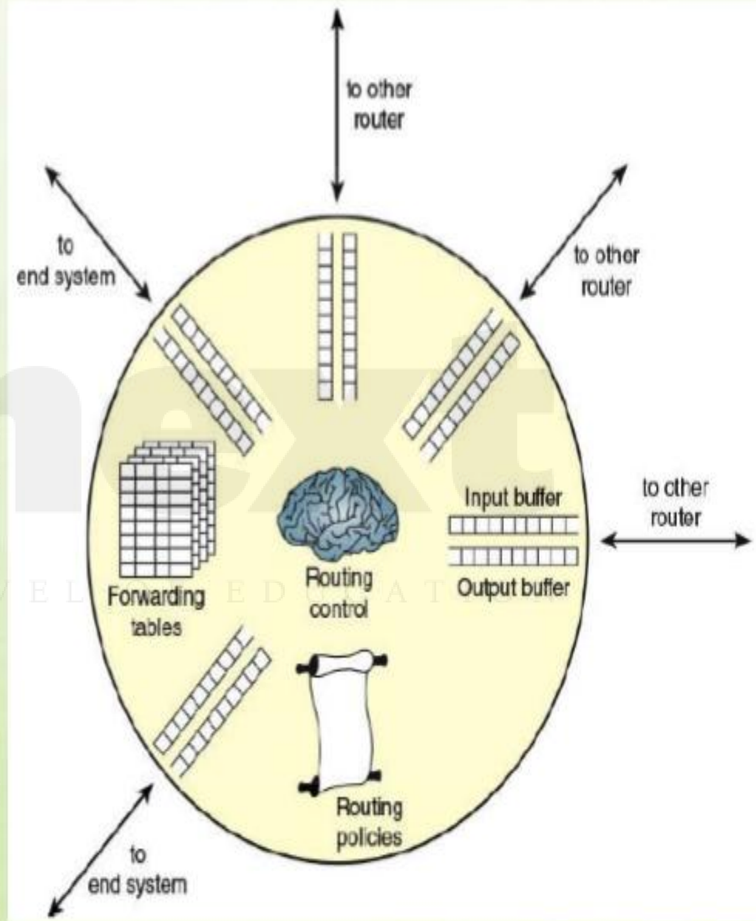
PART I MODERN NETWORKING

Chapter 2: Requirements and Technology

2.4 ROUTING

Elements of a Router

- The router examines each incoming packet, makes a routing decision based on the forwarding tables, and then moves the packet to the appropriate output buffer.
- Packets queued for output are transmitted as rapidly as possible.
- Each output queue can be operated as a simple first-in, first-out (FIFO) queue.
- More commonly, a more complex queuing discipline is used, to take into account the relative priority of the queued packets.
- A set of routing policies may also influence the construction of the forwarding tables and how various packets are to be treated.
- Policies may determine routing not just on the destination address but other factors, such as source address, packet size, and protocol of the payload.
- The final element shown in Figure is a routing control function.
- This function includes execution of routing protocols, adaptive maintenance of the routing tables, and supervising congestion control policies.



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